

Name: _____



Checking and securing understanding of plotting coordinates with technology

Task A: Plotting sequences and rules using technology

1) Use the table function in Desmos to generate the first ten terms of the arithmetic sequence $20 - 4n$

Once plotted resize your axes so that all ten terms are visible.

2) Use the table function to generate the first ten terms of the arithmetic sequence $2.65n - 17.5$

Use your graphical representation to identify the first positive term.

3) Use Desmos to explore if the sequences $5n - 40$ and $56 - 3n$ share a common term.

4) Use Desmos to graphically represent the rules $y = 6\frac{1}{2} - \frac{1}{4}x$ and $y = 3.26x + 18.7$

Identify the quadrant in which they intersect.

Task B: Identifying if points fit a rule

1) Use Desmos to find which coordinate does not fit the same rule as the others.

(-5, 23)

(1, 5)

(-1, -11)

(0, 8)

(4, 4)

(-3, 17)

(6, -10)

2) Which coordinates fit the rule $y = \frac{7}{4}x - \frac{2}{3}$?

$\left(\frac{1}{2}, \frac{5}{24}\right)$

$\left(2, \frac{17}{6}\right)$

$\left(-2, -\frac{25}{6}\right)$

$\left(\frac{2}{3}, 2\right)$



3) Use Desmos to graphically represent the rules

$$y = 3x - 13 \text{ and } y = 17 - 2x$$

Identify which of these three coordinates fits both rules.

(5, 7)

(6, 5)

(7, 8)